Edge Micropayments, Inc.
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October 14, 2025

Ms. Ruth Okyere
Opal

Dear Ruth,

Based on our conversations we present below our proposal for Edge Micropayments' design and development services for Opal's nurse-worn medical assistant and its integration with the Edge Micropayment Platform for Integrated Commerce (EMPIC™).

Introduction to Edge Micropayments and EMPIC

Edge Micropayments, Inc. designs and operates EMPIC, a low-latency, non-custodial payment gateway optimized for machine-to-machine and human-in-the-loop micro-transactions. EMPIC combines secure device identity (DIDs), escrow-backed settlement in stablecoins (e.g., USDC on L2), and programmable split-payouts to reduce transaction costs by orders of magnitude while improving auditability and cash flow.

What we provide

Device & client engineering: Wearable client design (hardware integration, firmware, edge app), secure audio capture/streaming, session UX, SSO integration.

Provider/Gateway engineering: LLM/ASR/TTS gateway, metering, routing, retrieval-augmented Q&A against hospital policies, multilingual translation.

EMPIC integration: Consumer & Provider SDKs, non-custodial escrow, split payouts, automated rollups, Stripe on/off-ramp orchestration to hospital treasury.

Security & compliance: HIPAA/PHI-safe data flows, least-privilege access, signed receipts, audit hooks.

Phase 1 — MVP: Wearable assistant + LLM integration (policy Q&A + translation)

Scope

Wearable client (Opal): Lapel device/companion app with push-to-talk, wake-word or tap UI, secure audio capture, and return audio streaming.

Opal Device gateway ("Provider" side): Low-latency pipeline for ASR → LLM → TTS, real-time partials, and metering counters (ASR seconds, tokens, TTS chars).

Policy Q&A & translation: LLM(s) tuned on Mount Sinai policy/SOP corpus (through retrieval), plus short-form language translation for bedside use.

Hosting model: LLM/ASR/TTS hosted on a secure server either (a) within hospital IT boundaries (on-prem/VPC), or (b) in a hardened virtual cloud platform (VCP) connected via mTLS and IP allow-lists.

Voice & text tooling: Where appropriate, integrate via API with vendors such as 11ElevenLabs (https://elevenlabs.io) for AI voice and text processing to accelerate MVP.

Identity & access: SSO/SAML, device enrollment, signed session receipts; PHI-safe boundary where audio/text remains in the hospital perimeter when required.

Deliverables

- Wearable client application (buildable image + configuration).
- Edge gateway service with endpoints for ASR/LLM/TTS, session logs, and metering.
- Data connectors for Mount Sinai policy retrieval and translation models.
- Security & compliance checklist (HIPAA/PHI scope, logging, retention).
- Pilot runbook: deployment, monitoring, incident handling, and rollback.

Standards & compliance

All Phase-1 work will be performed in conformance with HIPAA requirements and industry best practices for handling PHI (encryption in transit with TLS 1.2+, optional mTLS, strict scoping, and retention controls).

Phase 2 — EMPIC Payment Integration: Escrow, Metering & Split-Payouts

Objective

Convert the functional MVP into a commercially viable, escrow-backed service with automated settlement and finance-clean reporting inside the hospital.

Why EMPIC?

- Ultra-low fees on micro-transactions via rollups and net-settlement.
- Non-custodial escrow: Hospital controls keys; smart contracts enforce release/splits—reduced regulatory surface.
- Faster cash cycles: Escrow release on fulfillment; optional scheduled off-ramp to treasury.
- CapEx to OpEx shift: Treats inference/translation as granular operating expense with per-session caps.
- Better records: Signed, tamper-evident receipts for every interaction; clean GL exports.

What we implement

EMPIC SDK on the wearable (Consumer): DID bootstrap, ephemeral cryptographic token (ECT) per session, open/close escrow caps (USDC on L2), display remaining session balance.

EMPIC SDK at the gateway (Provider): Price registration, escrow verification, precise metering, and release intent submission; split-payout to the LLM service and EMPIC rail-fee.

Treasury operations (e.g., Stripe integration):

On-ramp: Hospital fiat → Stripe → USDC → hospital wallets (consumer/provider pools).

Off-ramp: Wallet → USDC redeem → Stripe → hospital bank account (optional netting schedule).

Commercial configuration:

- Subscription (per clinician/department)

- Pay-as-you-go (metered minutes/tokens)
- Hybrid (base subscription + metered overage)

Illustrative fee comparison (typical micro-transaction)

Payment rail	Example fee on $1.00 txn	Notes
Traditional card (e.g., 2.9% + $0.30)	$0.329	Fixed per-transaction fee dominates small amounts.
ACH (varies by processor)	~$0.26–$1.50	Slower settlement, return risks, manual exception handling.
EMPIC (escrow + rollup)	**~$0.0002** (≈0.02%)	Illustrative bps-based rail fee; on-chain costs amortized in batch.

Figures are illustrative for planning; final pricing subject to hospital configuration, volumes, and selected L2.

Additional benefits

- Shorter payment cycles via programmable release.
- Predictable budgeting with escrow caps and per-unit pricing.
- Automated reconciliation and GL-ready statements.
- Lower chargeback exposure (service proof linked to release).
- Auditability: cryptographic receipts tied to device DID, model version, and retrieval set.

Integration roles

EMPIC Consumer SDK (wearable): session auth (DID/ECT), open/close escrow, stream usage hints, present remaining balance.

EMPIC Provider SDK (gateway): verify escrow, meter ASR/LLM/TTS precisely, submit release, receive split-payout.

Stripe: on/off-ramp to hospital treasury (optional netting).

Standard Terms

IP Ownership & License to Deliverables (Interim)

Background IP: Each party retains its own pre-existing IP and tools (Opal: device concepts/brand; Edge: EMPIC platform, SDKs, contracts).

Wearable & Client UX: Opal owns the wearable hardware design, device firmware specific to Opal, and client UX produced under this SOW.

EMPIC Platform: Edge owns EMPIC (platform, SDKs, contracts) and related tooling.

Integration Software (Server-Side): TBD. Ownership and licensing for server-side adapters/APIs connecting Opal and EMPIC will be finalized in a short IP Allocation Addendum. Options include: co-ownership with cross-licenses, single-party ownership with a perpetual license to the other, or module-by-module allocation.

Interim Rights (Pilot): Until the addendum is signed:

Opal has a non-exclusive license to use EMPIC SDKs/runtime to operate the MVP and pilot.

Edge has a limited license to use Opal's device/client artifacts only to integrate, test, and support EMPIC flows.

Third-Party/Open Source: Any third-party or OSS components will be listed with applicable licenses.

Next Step: The parties will finalize ownership, licenses, and any field-of-use limits in the IP Allocation Addendum before Phase 2 begins.

Confidentiality & PHI

Mutual NDA: The parties will protect each other's confidential information with the same care used for their own, and at least a reasonable standard of care.

PHI Safeguards: For any Protected Health Information (PHI) processed, the parties will implement least-privilege access, encryption in transit (TLS 1.2+), secure key management,

and retention/disposal consistent with the SOW and applicable law. If required, a Business Associate Agreement (BAA) will be executed.

Warranties & Disclaimers

Services Warranty: Edge warrants that services will be performed in a professional and workmanlike manner consistent with industry standards.

Third-Party Components: No warranty is provided for third-party software, models, or services (e.g., LLMs, ASR/TTS vendors); such items are provided “as is.”

MVP Scope: The MVP is for pilot validation only and is not intended for clinical diagnosis or decision-making without appropriate clinical oversight and approvals.

Indemnification

Mutual IP Indemnity: Each party will indemnify, defend, and hold the other harmless from third-party claims alleging that its in-scope deliverables infringe IP rights, except where claims arise from misuse, modifications, or combinations not authorized by the indemnifying party.

Process: Indemnification is conditioned on prompt notice, reasonable cooperation, and the indemnifying party’s sole control of defense and settlement (no admission of liability or injunctive relief without consent).

Limitation of Liability

Cap: Except for breach of confidentiality, IP infringement indemnity, or willful misconduct, each party’s aggregate liability under the SOW is capped at the fees paid or payable for the services giving rise to the claim.

Exclusions: No liability for indirect, incidental, special, consequential, or punitive damages, including lost profits or revenue, even if advised of the possibility.

Compliance

The parties will comply with applicable laws, regulations, and hospital policies, including HIPAA and other privacy/security requirements identified in the SOW. Data flows and hosting locations will conform to the agreed PHI handling design.

Export Control & Sanctions Compliance

Each party will comply with all applicable U.S. and international export and sanctions laws, including the EAR (15 C.F.R. Parts 730–774), ITAR (22 C.F.R. Parts 120–130), and OFAC programs. The parties will not export, re-export, transfer, or provide access to services, software, technical data, or PHI to prohibited jurisdictions or restricted parties.

Licenses & Approvals: Each party is responsible for obtaining any required government licenses for its exports/re-exports.

ITAR-Controlled Data: No ITAR-controlled information will be introduced without prior written consent and appropriate controls.

Classification: For EMPIC SDKs and related materials, each party will classify its items (e.g., EAR99 or specific ECCN) and notify the other party where classification affects hosting, support personnel, or permitted end-users.

Change in Law: If a legal change renders performance non-compliant, the affected party may suspend impacted obligations on written notice; both parties will cooperate to implement a compliant alternative promptly.

Remedies: Breach of this section is a material breach and includes a duty to indemnify the non-breaching party for fines and penalties arising from the breaching party's non-compliance.

Force Majeure

Definition

Except for accrued payment obligations, neither party is liable for delay or failure to perform to the extent caused by a Force Majeure Event—events beyond reasonable control—including acts of God; natural disasters; epidemics/pandemics; war, terrorism, civil unrest; government actions or changes in law; nationwide or regional carrier/utility failures; widespread cloud or data-center outages; significant cyberattacks (including DDoS) not attributable to a failure to maintain industry-standard controls; non-party labor disputes; and similar events.

Process & Mitigation

Notice: The affected party will notify the other within five (5) business days of awareness, describing the nature, expected duration, and mitigation steps.

Efforts: Performance is suspended only to the impacted extent; the affected party will use commercially reasonable efforts to mitigate and resume performance promptly, including workarounds or alternate providers.

Schedule Relief: Milestones and deadlines are extended for the period of delay caused by the event.

Extended Events & Termination

If a Force Majeure Event continues for thirty (30) consecutive days (or sixty (60) days in the aggregate in any ninety-day period) and materially prevents performance of critical obligations, either party may terminate the affected Order/SOW on written notice. The customer will pay for services actually performed through the effective date, and the provider will refund any prepaid, unused fees for the terminated portion.

Security & PHI

Despite a Force Majeure Event, each party will maintain commercially reasonable security measures to protect PHI and confidential information and will promptly notify the other party of any security incident affecting such data, consistent with law and the agreement's incident-response obligations.

Commitment

Phase 1

Team: Two dedicated contributors (engineering + strategic design)

Effort: 80 hours (design & engineering) + 16 hours (strategic consulting)

Duration: ~4 weeks

Fees: Billed at Edge Micropayments' standard rates; final amount will reflect actual hours expended against the stated allocations (rate card to be attached to the SOW).

Phase 2

Phase-2 pricing will be scoped after Phase-1 acceptance, reflecting subscription, metered usage, or hybrid configuration and projected volumes.

Next Steps

We're excited to partner with Opal to deliver a safe, fast, and truly useful nurse-worn assistant—and to prove a payment and settlement model that is both hospital-friendly and economically sustainable. We look forward to your feedback on this proposal and to moving ahead with mutually understood and agreed upon changes.

Sincerely,

Ted Schwartz
Chief Executive Officer
Edge Micropayments, Inc.